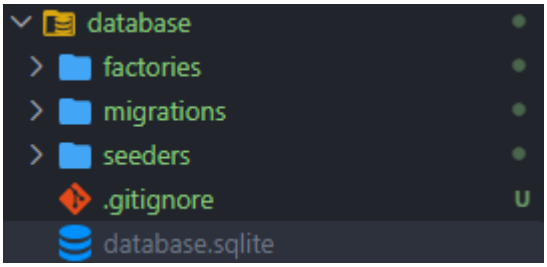
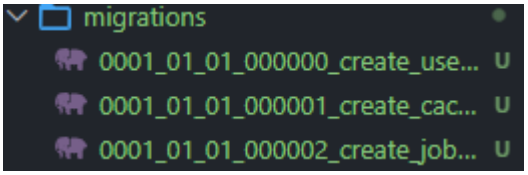
5.1 BDD Config

estructura de directorios de la bbdd



carpeta migraciones



conexiones en .env

```
DB_CONNECTION=sqlite
# DB_HOST=127.0.0.1
# DB_PORT=3306
# DB_DATABASE=laravel
# DB_USERNAME=root
# DB_PASSWORD=
```

funcion up()

define los cambios a aplicar

```
     * Run the migrations.
     */
    public function up(): void
    {
        Schema::create('users', function (Blueprint
```

funcion down()

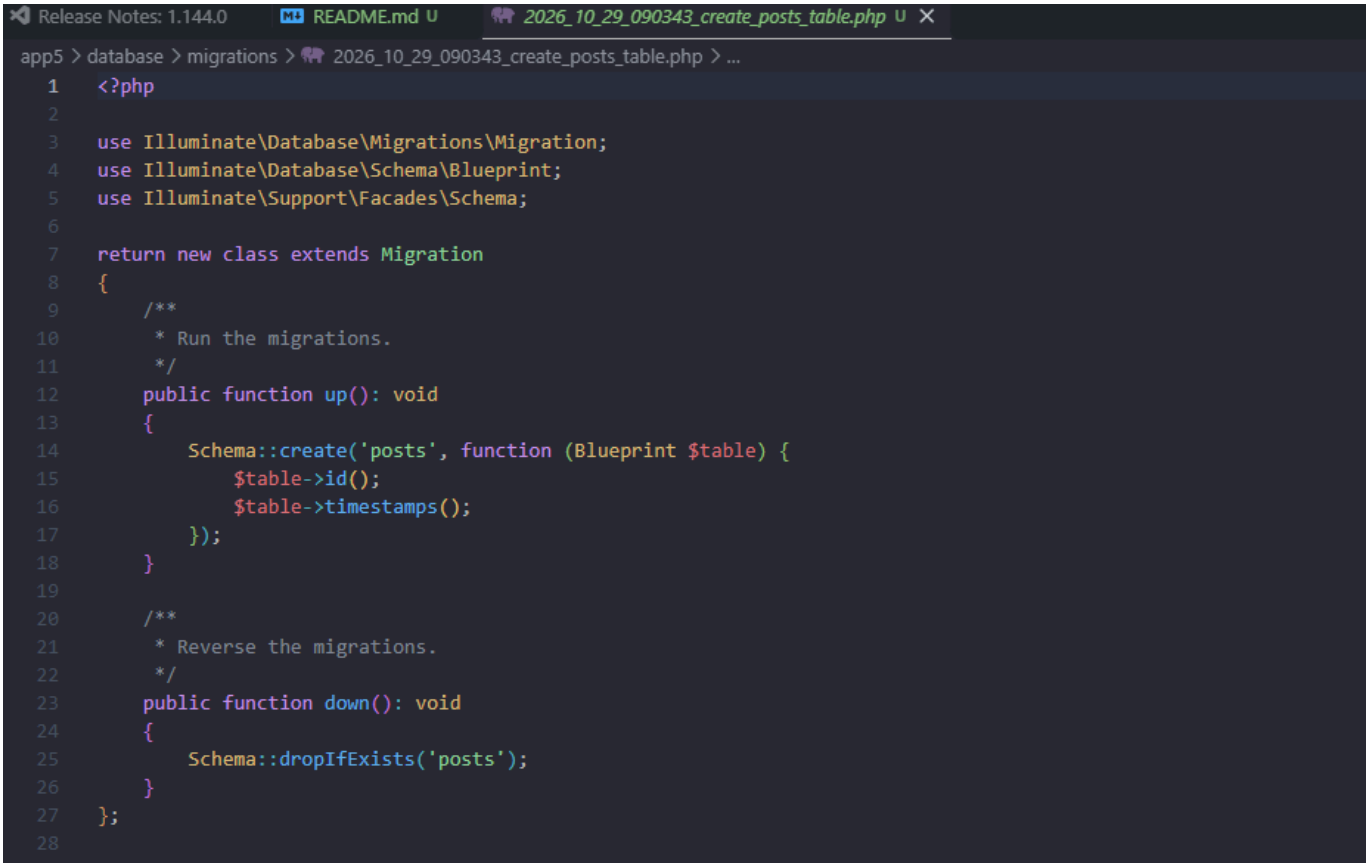
define cómo deshacer esos cambios

```
    /**
     * Reverse the migrations.
     */
    public function down(): void
    {
        Schema::dropIfExists('users');
        Schema::dropIfExists('password_reset_t
        Schema::dropIfExists('sessions');
```

reacion de la tabla usuario

```php
Schema::create('users', function (Blueprint $table) {
    $table->id();
    $table->string('name');
    $table->string('email')->unique();
    $table->timestamp('email_verified_at')->nullable();
    $table->string('password');
    $table->rememberToken();
    $table->timestamps();
```

Crear migracion



Ejecutar migraciones



Revertir migraciones

```
PS C:\xampp\htdocs\dwesAntonio\DWES_AntonioJesusLuque\UD6-Laravel\S5\app5> php artisan migrate:rollback

  INFO  Rolling back migrations.

  2026_10_29_090343_create_posts_table ..... 35.99ms DONE

PS C:\xampp\htdocs\dwesAntonio\DWES_AntonioJesusLuque\UD6-Laravel\S5\app5>
```

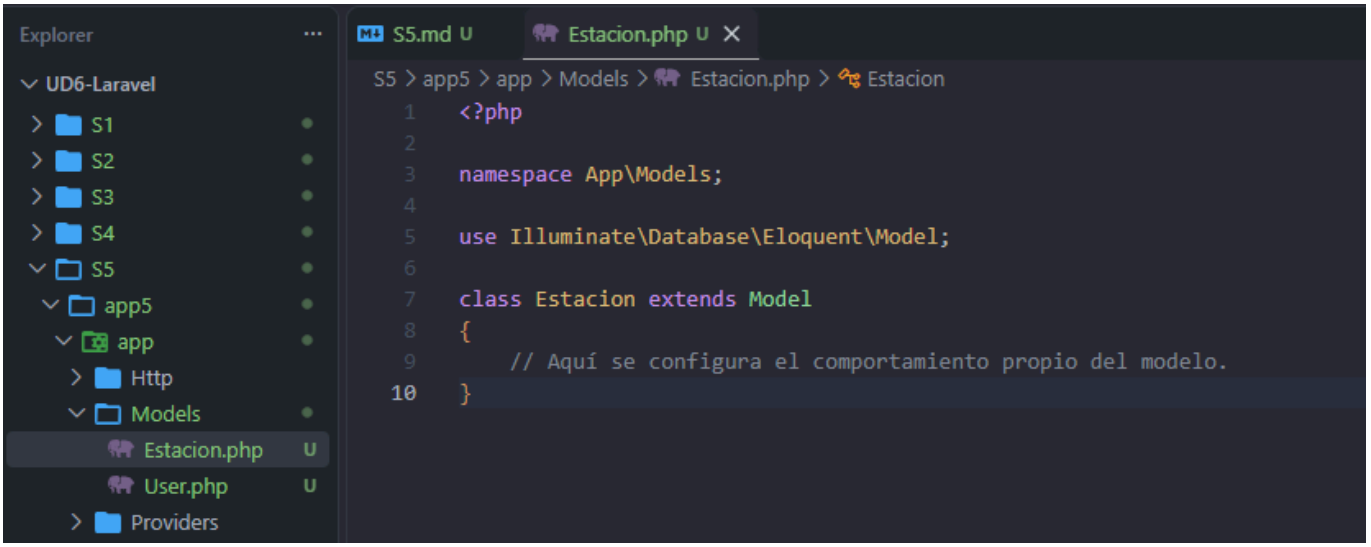
Crear modelos

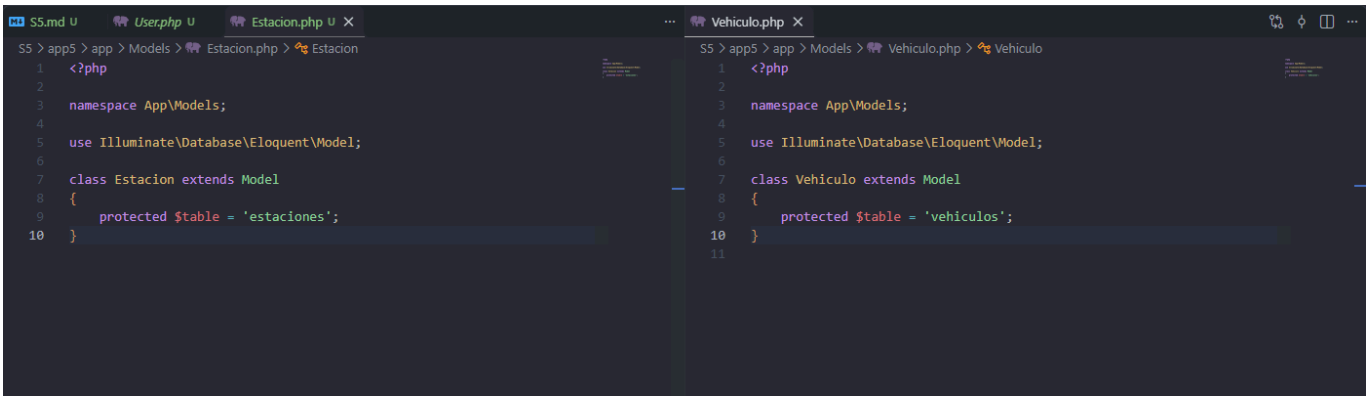
```
Problems  Output  Debug Console  Terminal  Ports

PS C:\xampp\htdocs\dwesAntonio\DWES_AntonioJesusLuque\UD6-Laravel\S5\app5> php artisan make:model Estacion

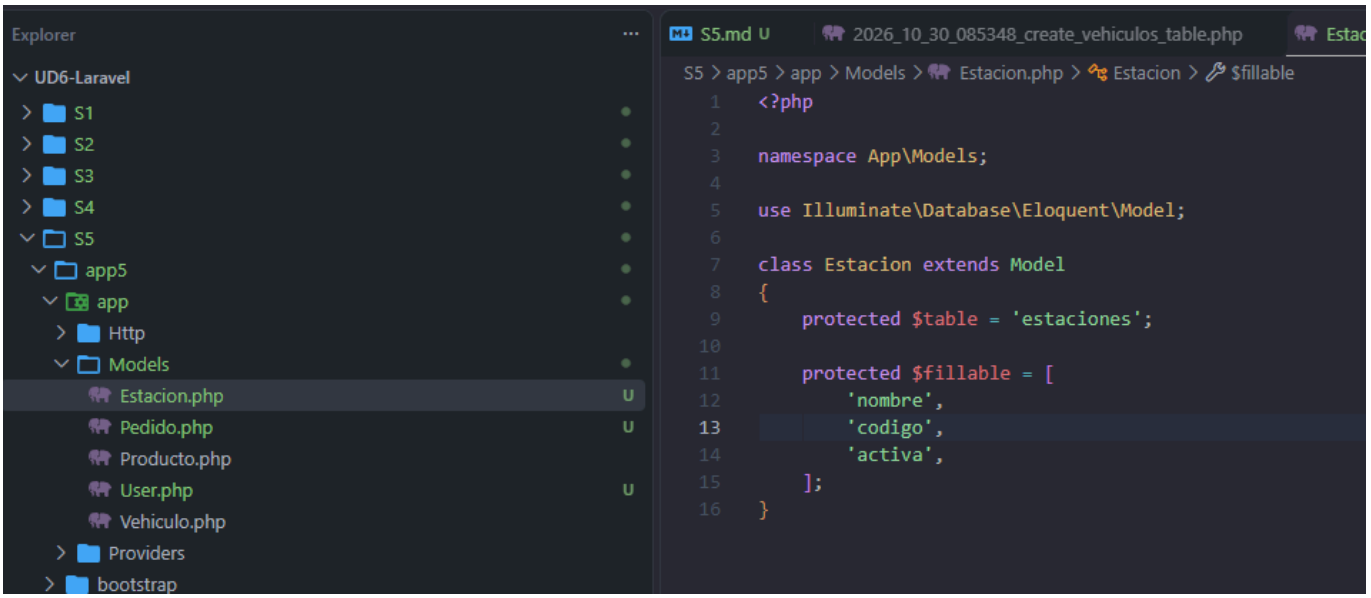
  INFO  Model [C:\xampp\htdocs\dwesAntonio\DWES_AntonioJesusLuque\UD6-Laravel\S5\app5\app\Models\Estacion.php] created successfully.

PS C:\xampp\htdocs\dwesAntonio\DWES_AntonioJesusLuque\UD6-Laravel\S5\app5>
```

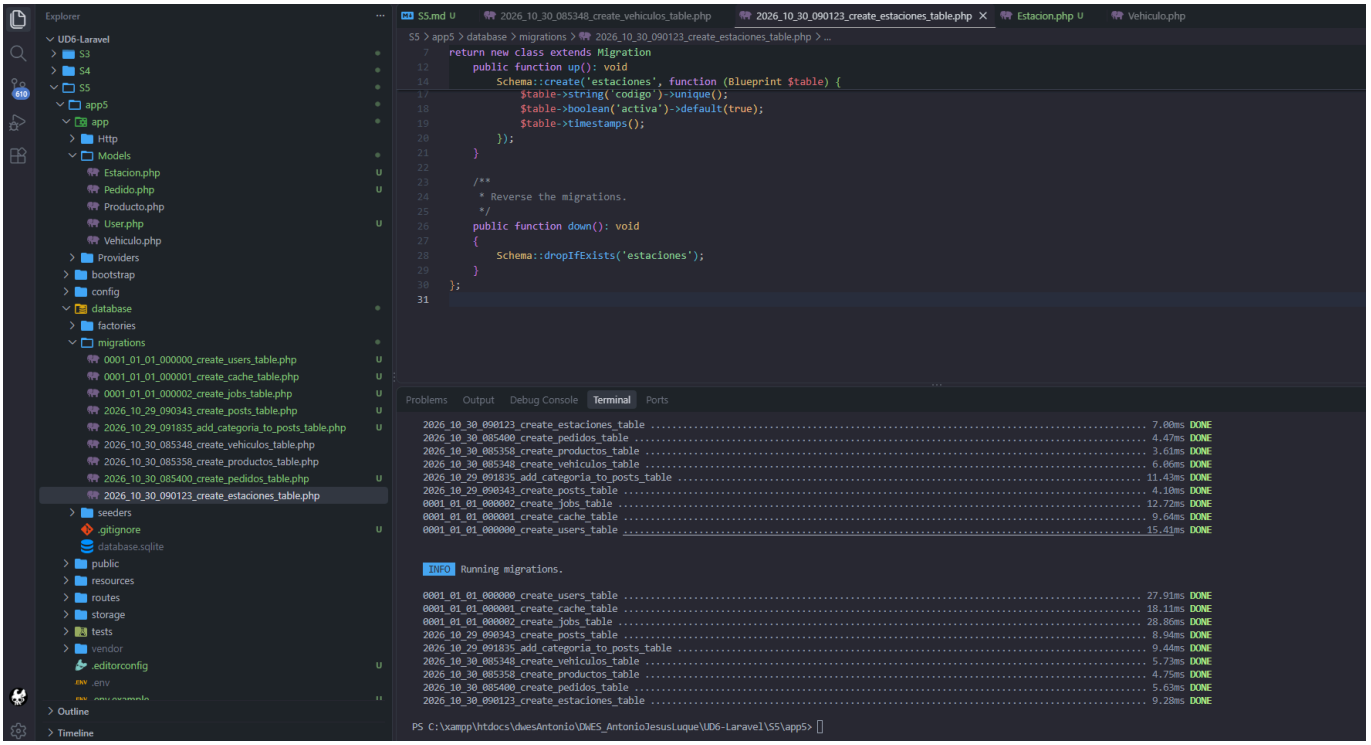




Por seguridad, Eloquent protege los modelos frente a la asignación masiva. Hay que indicar qué campos se pueden asignar con `$fillable`



En `Vehiculo` se puede autorizar la clave foránea y los datos propios del vehículo



Solucion app5

Estacion

```php
<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Model;
use Illuminate\Database\Eloquent\Relations\HasMany;

class Estacion extends Model
```

```
{
    //
    protected $table = 'estaciones';

    protected $fillable = [
        'nombre',
        'codigo',
        'activa',
    ];

    public function vehiculos(): HasMany
    {
        return $this->hasMany(Vehiculo::class);
    }
}
```

vehiculo

```
<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Model;
use Illuminate\Database\Eloquent\Relations\BelongsTo;

class Vehiculo extends Model
{
    //
    // protected $table = 'vehiculos';
    // protected $primaryKey = 'codigo';

    // Define qué campos permites registrar masivamente
    protected $fillable = [
        'estacion_id',  // Foreign key to the Estacion model
        'modelo',
        'tipo',
        'disponible',
    ];

    public function estacion(): BelongsTo
    {
        return $this->belongsTo(Estacion::class);
    }
}
```

migracion create vehiculos

```
<?php
```

```php
use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
     */
    public function up(): void
    {
        Schema::create('vehiculos', function (Blueprint $table) {
            $table->id();
            $table->foreignId('estacion_id')
                ->constrained('estaciones')
                ->cascadeOnDelete();
            $table->string('modelo', 100);
            $table->string('tipo', 30);
            $table->boolean('disponible')->default(true);
            $table->timestamps();
        });
    }

    /**
     * Reverse the migrations.
     */
    public function down(): void
    {
        Schema::dropIfExists('vehiculos');
    }
};
```

migracion create estaciones

```php
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
     */
    public function up(): void
    {
        Schema::create('estaciones', function (Blueprint $table) {
            $table->id();
            $table->string('nombre');
            $table->string('codigo')->unique();
            $table->boolean('activa')->default(true);
```

```
            $table->timestamps();
        });
    }

    /**
     * Reverse the migrations.
     */
    public function down(): void
    {
        Schema::dropIfExists('estaciones');
    }
};
```

tinker

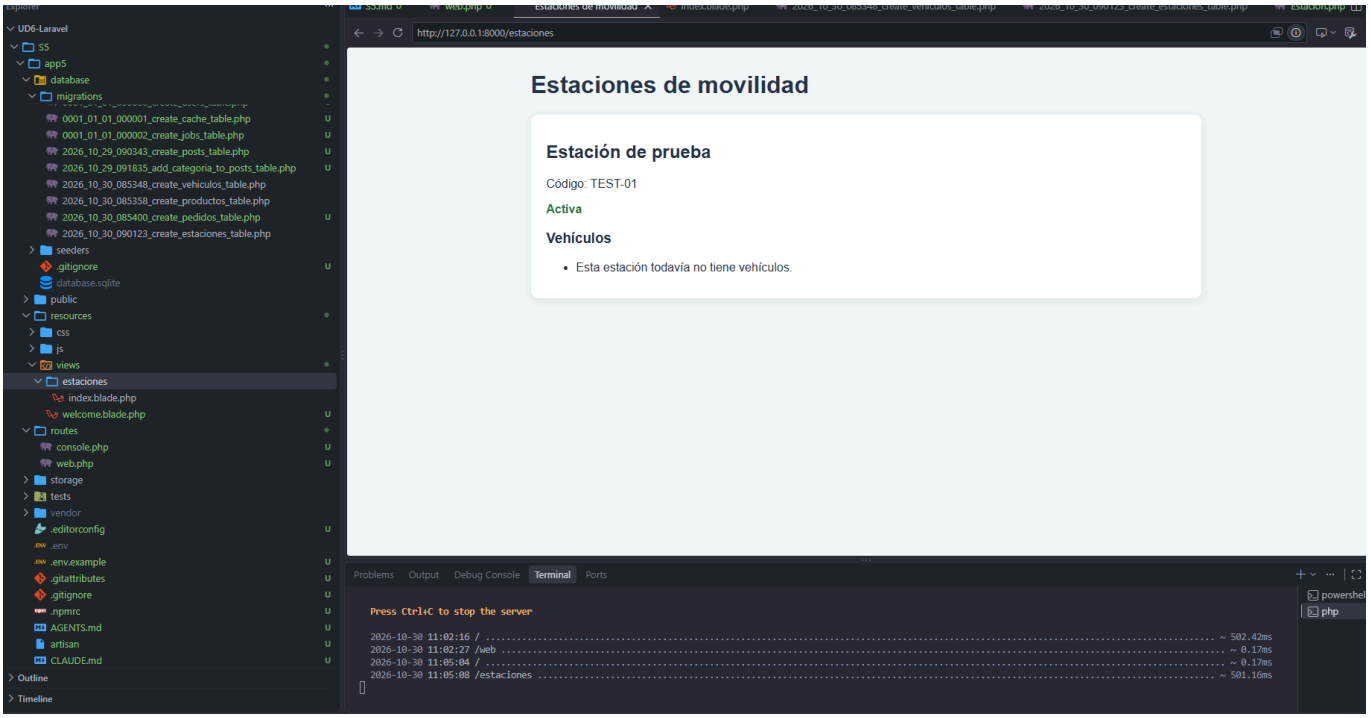
```
PS C:\xampp\htdocs\dwesAntonio\DWES_AntonioJesusLuque\UD6-Laravel\S5\app5> php artisan tinker
Psy Shell v0.12.24 (PHP 8.5.12 — cli) by Justin Hileman
New PHP manual is available (latest: 3.1.4). Update with `doc --update-manual`
> $estacion = App\Models\Estacion::create([
.     'nombre' => 'Estación de prueba',
.     'codigo' => 'TEST-01',
.     'activa' => true,
. ]);

= App\Models\Estacion {#7954
    nombre: "Estación de prueba",
    codigo: "TEST-01",
    activa: true,
    updated_at: "2026-10-30 09:56:44",
    created_at: "2026-10-30 09:56:44",
    id: 1,
  }

>
> $vehiculo = $estacion->vehiculos()->create([
.     'modelo' => 'Bicicleta urbana',
.     'tipo' => 'bicicleta',
.     'disponible' => true,
.     'autonomia_km' => 65.50,
. ]);
```

mostrar los datos en una vista

Después de crear y probar los modelos, presenta los datos en una página web. La consulta se hará desde una función anónima





show.blade.php

